UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2019/2020

SEPTEMBER EXAMINATION

UECS3253/UECS3463 WIRELESS APPLICATION DEVELOPMENT

TUESDAY, 10 SEPTEMBER 2019 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF SCIENCE (HONS) SOFTWARE ENGINEERING
BACHELOR OF ENGINEERING (HONOURS) ELECTRONICS (COMPUTER NETWORKING)

Instructions to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Section A: Answer **ALL** questions.

Section B: Answer **ONE (1)** question only.

Each question carries 25 marks.

Note: Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the FIRST question in Section B in the order the candidate submits the answer.

Answer the questions in the answer booklet provided.

UECS3253/UECS3463 WIRELESS APPLICATION DEVELOPMENT

SECTION A: Answer **ALL THREE (3)** questions in this section.

Q1. (a) Mobile apps can be classified into *native apps*, *mobile web apps* and *hybrid apps*. Explain **THREE (3)** differences between *native apps* and *mobile web apps*. (12 marks)

(b) React Native is a framework currently used to construct mobile apps for both Android and iOS platforms. Imagine a possible future where each and every Chinese smartphone manufacturers decide to release their own operating systems to power their smartphones and these smartphones constitute a significant market share worldwide.

Explain the advantages of developing mobile apps using React Native framework in the abovementioned scenario. (13 marks)

[Total : 25 marks]

Q2. Table 2.1 shows the sample data of a table named **fbpages** in an SQLite database named **catfanclubsDb** that is used in a mobile app providing a listing of all cat fan clubs on Facebook.

id	fbid	name
1	HappyCatsOnline	Happy Cats
2	Gresta.meow	Cats Universe
3	lovecattom	Love Cat Tom
4	KittenHappiness	Kitten Happiness

Table 2.1: Sample data of *fbpages* table

(a) Data of Facebook Pages are downloaded by the mobile app to be inserted into the abovementioned **fbpages** table upon first launch of the app. The data is in JSON format and is available in a variable named **catPages**. Figure 2.2 shows the sample data in JSON format.

```
[
    {
        "fbid": "HappyCatsOnline",
        "name": "Happy Cats"
    },
    {
        "fbid": "Gresta.meow",
        "name": "Cats Universe"
    },
    {
        "fbid": "lovecattom",
        "name": "Love Cat Tom"
    },
    {
        "fbid": "KittenHappiness",
        "name": "Kitten Happiness"
    }
]
```

Figure 2.2: Sample JSON data

Construct code segment to insert new rows to the **fbpages** table using the JSON data in the **catPages** variable. (8 marks)

Q2. (Continued)

(b) Updates to the data about the Facebook pages are downloaded by the app to be updated into the **fbpages** table. The data is in JSON format as shown in the sample in Figure 2.2 and is available in a variable named **pagesUpdate**.

Construct code segment to insert or update the rows in **fbpages** using the JSON data in the **pagesUpdate** variable. If the **fbid** is already exist, perform an update. Otherwise, insert a new row. (11 marks)

(c) A screen of the mobile app displays a **FlatList** listing the names of all Facebook Pages.

Constuct code segment to query the **fbpages** table and prepare the data as a JavaScript object that is suitable to populate the **FlatList** with the names of the Facebook Pages. (6 marks)

[Total : 25 marks]

Q3. The **Settings** screen of a React Native mobile app allows the user to set various preferences and personalizations that will be applied. The user preferences are persisted locally on the device storage using **AsyncStorage**.

When the app is opened, the values will be loaded from **AsyncStorage** and set to the corresponding states where the key names match the respective state names.

When the user opens the **Settings** screen and makes changes to a preference, the modified preference will be saved and applied immediately.

Table 3.1 lists the available settings and their respective key names and sample values for this app.

Key	Sample Value
textColor	#000000
backgroundColor	#cfcfcf
showNotifications	true
playSoundEffects	false
playMusic	false
useLocationInfo	true
preferWifi	true
doNotUseMobileData	false
showNearby	true
fontFamily	Arial, Sans-serif
fontSize	medium
fontStyle	normal
mediaAutoDownload	false
mediaAutoPlay	false
language	en-US

Table 3.1: Available Settings

(a) Construct the code segment to load the respective values of all keys in Table 3.1 and set the states accordingly. Display an alert with an appropriate message if an error occurs while retrieving the values. (9 marks)

Q3. (Continued)

(b) Construct the code segment to save a single preference using value from the state to the corresponding key in **AsyncStorage**. Your code should be general and works on all preference keys shown in Table 3.1.

Display an alert with an appropriate message if an error occurs while saving the value. (7 marks)

(c) The **Settings** screen of the mobile app also has a **Save All & Apply** button. Clicking on this button will cause all the preference values shown in Table 3.1 to be saved into **AsyncStorage** from the corresponding state.

Construct the code segment to implement the abovementioned requirements.

Display an alert with an appropriate message if an error occurs while saving all the values. (9 marks)

[Total : 25 marks]

SECTION B: Answer **ONLY ONE (1)** question from this section.

Q4. Yoyaku is a hotel booking platform that allows a user to book for hotels and motels in many destinations worldwide. The platform allows users to book for accomodation with the use of a mobile app.

(a) A customer who wishes to use Yoyaku is required to register as a user.

Figure 4.1 shows a sample JSON data that is sent by the app to the web server during registration while Figure 4.2 shows a sample response from the web server.

```
{
    "email": "lili@imail.com",
    "phone": "+60123456789",
    "password": "EbV6^evULU",
    "name": "Lim Li Li",
    "gender": "F",
    "countryOfResidence": "AU",
    "nationality": "MY"
}
```

Figure 4.1: Sample JSON data for user registration

```
{
    "id": 1019,
    "email": "lili@imail.com",
    "registered": true
}
```

Figure 4.2: Sample response

Constuct code segment to send the JSON data in Figure 4.1 to a web server at the URL **https://yoyaku.travel/api/register** using HTTP POST method. Assume that the JSON data is already set to a variable named **user**.

If the user is successfully registered, navigate the user to another screen using a route named **SearchHotel**, passing the **id** as a route parameter. Otherwise, display an alert with an appropriate error message.

Should any error occurs when attempting to connect and communicate with the web server, display an alert with an appropriate error message. (13 marks)

Q4. (Continued)

(b) A user searches for available accomodation by providing keywords and various search parameters. The app sends the user's search criteria in JSON format as shown in the sample in Figure 4.3 to the web server using HTTP POST method at the URL **https://yoyaku.travel/api/booking/search**. The server responds with JSON data containing the available hotels as shown in Figure 4.4.

```
{
    "keywords": "Penang",
    "checkIn": 1549324800,
    "checkOut": 1549670400,
    "adults": 2,
    "children": 3
}
```

Figure 4.3: Sample JSON data sent to server

```
[
    {
        "id": 7543,
        "name": "Starway Hotel",
        "city": "George Town",
        "state": "Penang",
        "country": "MY",
        "stars": 4,
        "rating": 8.5
    },
    {
        "id": 8681,
        "name": "Vista Hotel",
        "city": "Bayan Lepas",
        "state": "Penang",
        "country": "MY",
        "stars": 3,
        "rating": 8.6
    }
]
```

Figure 4.4: Sample JSON data of search results

Construct code segment to send the JSON data in Figure 4.3 to the web server. Upon receiving the JSON response as shown in Figure 4.4, the app navigates the user to another screen using a route named **SearchResults**, passing the JSON response in Figure 4.4 using a route parameter named **results**. Otherwise, navigate the user to another screen using a route named **NoResults**. Assume that the data in Figure 4.3 has been set in a variable named **searchParams**.

(12 marks)

[Total : 25 marks]

Q5. MyTVGuide is a mobile app that provides listing of TV programmes aired on Malaysian free-to-air TV channels. This app is required to make a *socket* connection using TCP to a server at **mytvguide.my:8510** to send and receive data between the app and the server.

(a) Users of the app are required to specify the TV channel and date to display the listing of TV programmes. The app sends the user's selection to the server as shown in the sample JSON data in Figure 5.1. The server responds to the app with the data of the listing in JSON data as shown in the sample in Figure 5.2.

```
{
    "channel": "tv2",
    "date": "2019-02-05"
}
```

Figure 5.1: Sample data from app to server

```
{
    "06:00": "Motivation",
    "07:00": "Health & Wealth",
    "08:00": "Looney Tunes",
    "09:00": "Hong Bao Na Lai",
    "10:00": "Zodiac 2019",
    "10:30": "CNY Telemovie: Tahun Baru di Kuching",
    "12:00": "Mandarin News",
    "12:30": "News On 2",
    "13:00": "CNY Movie - Chicken & Duck Talk",
    "15:00": "CNY Movie - All's Well Ends Well 97",
    "17:00": "CCTV Chinese New Year Special",
    "18:30": "Zodiac 2019",
    "19:00": "News On 2",
    "19:30": "Tamil News",
    "20:00": "Mandarin News",
    "20:30": "Pesta Hong Bao 2019",
    "22:30": "CNY Movie: Kung Fu Yoga",
    "00:30": "CNY Telemovie: Tahun Baru di Kuching (repeat)",
    "02:30": "CCTV Chinese New Year Special (repeat)",
    "04:00": "Pesta Hong Bao 2019 (repeat)"
}
```

Figure 5.2: Sample listing data

Construct code segment to open a TCP connection to the **WebSocket** URL **mytvguide.my:8510/listing** using the **WebSocket** API to send the data as shown in Figure 5.1 to the server.

When the response as shown in Figure 5.2 is received, navigate the user to another screen using a route named **Listing**, supplying the response data using a route parameter named **listing**. Assume that the data in Figure 5.1 is already set in a variable named **selection**. (16 marks)

Q5. (Continued)

(b) Users of the mobile app can also report any errors or ommisions in the listing of TV programmes by filling up a user feedback form. Figure 5.3 shows a sample data in JSON format of a user feedback that will be sent from the app to the server. The server responds to the app by sending the integer **1** on success, or **0** on error. Upon receiving the response, the app displays an alert with an appropriate message.

```
{
    "name": "Chia Kim Hooi",
    "email": "chiakh@imail.com",
    "phone": "012-4049792",
    "subject": "Incorrect listing for 8TV",
    "message": "Wrong timing for CNY movie on 4 Feb 2019"
}
```

Figure 5.3: Sample feedback data

Construct code segment to send the feedback as shown in Figure 5.3 to the server using the **WebSocket** URL **mytvguide.my:8510/feedback** and handle the response accordingly. Assume that the data in Figure 5.3 is already set in a variable named **userFeedback**. (9 marks)

[Total : 25 marks]